

Institutional Design and Liquidity at Stock Exchanges around the World

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Abstract

The paper investigates the trading mechanism and other structural features of 51 stock exchanges and analyzes the impact of these institutional characteristics on liquidity measures such as closing bid-ask spreads, volatility and trading turnover. Exchange-design features such as narrower tick sizes, designated market makers, consolidated limit order books, hybrid trading mechanisms, automated trade execution, consolidated order-flow, and better shareholder rights are associated with lower spreads. These features also influence volatility and trading turnover, which in turn further affect spreads. Overall liquidity is highest on hybrid trading mechanisms compared to pure limit order books or quote-based dealer system because the former have two sources of liquidity. These results have important implications for investors' trading strategy, firms' listing strategy, and exchanges' organizational strategy.

Stock market trading is witnessing radical changes at the dawn of the new millennium. Rising globalization, deregulation of the financial sector, cross listings, and foreign portfolio investments have made the competition among exchanges greater than ever before. In addition, technological advancements in telecommunications and the Internet are modifying the basic business model of a stock exchange. In response to these developments, stock exchanges around the world have implemented many important institutional-design changes.

*Hybridization, automation, consolidation, transparency, decimalization, and demutualization*¹ have become the buzzwords in the stock market circles. Exchanges now employ an amazing array of trading mechanisms both in terms of who supplies liquidity and how the actual transactions are completed. Only a few exchanges now rely completely on dealer quotes or open-outcry method, which historically were the dominant methods of trading. Most exchanges now have pure electronic limit order book trading with no designated market makers. The source of liquidity is customers' limit orders which are matched instantly or put in a queue using a computer algorithm. However, other exchanges worldwide have either introduced or proposed "hybrid" trading systems combining consolidated electronic public order books and obligatory quotes by designated market makers. The NYSE is already such a hybrid system, NASDAQ is proposing the 'Supermontage' limit order book that will supplement the quotes from competitive dealers, and Germany introduced hybrid systems in 1997. Automated trading is another major trend of the stock markets. Numerous cost effective electronic communication networks (ECN) have quickly captured a considerable market share of trading volumes² in the United States (US). Over 80% of the world's exchanges have introduced some form of electronic-screen-based trading mechanism with automatic execution. Many of these exchanges have completely abolished the floor trading mechanism. Another

¹ Demutualization separates ownership from membership and can potentially promote profitable policies that trading members themselves may not implement due to conflict of interest.

² ECNs captured over 30% of NASDAQ trades and over 4% of NYSE trades within 3 years of their existence.

important institutional trend is consolidation of order-flow. A wave of mergers has occurred among the US regional exchanges (see Arnold et al 1999) and among the leading European exchanges. Indeed, exchanges of Amsterdam, Brussels and Paris merged in September 2000 to form "Euronext". Similarly, in the last decade, Canada and France respectively consolidated all equity trading to a single exchange. These changes imply that we have more consolidated markets than fragmented ones. We also saw the world's leading exchanges (NYSE, NASDAQ, and Toronto) move from fraction to decimals pricing, thus reducing relative tick size. Finally, scores of exchanges in Sweden, Finland, Denmark, the Netherlands, Italy, Australia, Singapore, Hong Kong, Canada, and other countries³ are now incorporated and, thus, demutualized.

Even though exchanges are striving to bring about sweeping changes in their institutional design and policies, not many studies dwell upon the efficacy of their trading mechanism and other structural features in an integrated framework. The goal of this study is to test the theoretical predictions about improvement in performance of exchanges due to institutional factors such as presence of designated market makers (Black 1995, Rock 1989 and Stoll 1998, Glosten 1989), trading mechanism⁴ (Viswanathan and Wang 2002, Glosten 1994, and Seppi 1997), order-flow fragmentation versus centralization (Biais 1993, Hamilton 1972), reduced ex-ante transparency of order-flow details (Madhavan 1996), replacement of trading floors with automated screen-based trading (Domowitz and Stiel 1999), reduced tick size (Seppi 1997), de-mutualization of ownership (Domowitz and Stiel 1999), and enforcement of insider trading laws (Bhattacharya and Spiegel 1991). In order to test these hypotheses, this paper compares the performance of exchanges on various measures namely, quoted bid-ask spreads, effective spreads, realized spreads, volatility, and trading turnover.

³ The NYSE and the NASDAQ are co-owned by broker-members, but they have de-mutualization plans for future.

⁴ Three types of trading mechanisms are analyzed – dealer emphasis systems, pure limit order book systems and hybrid exchanges that combine the features of both.

Many empirical studies have previously compared the performance of different exchanges within the US. Huang and Stoll (1996) and Bessembinder and Kaufman (1997) report significant differences in trading costs on the NYSE, a hybrid market, and NASDAQ, a dealer market. Venkataraman (2001) reports that execution costs on the Paris Bourse are higher than they are on the NYSE. However, such bi-lateral comparisons do not capture all possible exchange-designs. For example, neither the NYSE nor the NASDAQ uses a pure limit order book trading mechanism, and neither is incorporated⁵. Another limitation of such comparisons is that these exchanges differ in more than one institutional characteristic such as existence of a limit order book, provision of automatic trade execution, order-flow transparency, order-flow fragmentation, presence of a designated market maker and so on. As a result, the impact of individual institutional features cannot be disentangled. In contrast, this study's extensive coverage of 51 exchanges provides a wide-ranging cross-sectional variation in both performance and institutional design measures. This study can therefore extricate the incremental impact of each institutional feature on performance of an exchange.

There are only a handful of cross-border comparisons of exchanges around the world. Domowitz et al (2002) use a US global portfolio manager's proprietary panel data from 42 countries from 1996 to 1998 to analyze the interactions between cost, liquidity, and volatility. They find a significant cross-sectional variation in total trading cost and its composition. Perold and Sirri (1997) and Chiyachantana, Jain, Jiang and Wood (2004) also use order and execution data from a large US institutional asset manager to document significant cross-country variation in equity trading costs. However, these studies do not directly relate the differences in observed spreads to institutional characteristics.

⁵ At least at the time of these studies, NYSE was hybrid and NASDAQ a dealer based system. The addition of international exchanges in the sample such as the Paris Bourse, which is a pure limit order book and is incorporated allows us to examine a much more diverse set of exchange designs.

This study extends the literature in an important way: it investigates whether the performance differences have their roots in institutional differences across exchanges. Further, unlike the previous studies, which consider total cost to US investors, this study considers costs to local investors. This approach therefore provides cleaner cost estimates and establishes the relationships between cost and institutional structure. The cost estimates are not contaminated by differential treatment that foreign investors receive in most countries. Further, an important methodological difference between this paper and Domowitz et al (2002) and Perold and Sirri (1997) is that whereas the latter studies compute implicit cost by taking the difference between transaction price and an indexed price⁶, this study uses the actual quoted and effective spreads at the close of each day. Quoted spreads are computed as the difference between the closing ask price and the closing bid price divided by the bid-ask midpoint for 25 securities with the highest market capitalization on each of 51 stock exchanges. Percentage effective spreads are computed as twice the difference between actual transaction price and quote midpoint divided by quote midpoint at the close of each day. These are likely to be much more accurate representations of costs especially if the intra-day volatility in prices is high. Higher volatility could widen the gap between transaction prices and indexed prices even though the actual spreads at any given point may be low⁷. To the best of our knowledge this is the first study that documents and uses the firm level transaction costs in 48 different countries to understand the effectiveness of institutional organization of exchanges. Although the highly fluid nature of both institutional-design and trading costs and a plethora of confounding events around the world make it very difficult to estimate accurate point estimates of these relationships, the study presents some general patterns and effects of stock-exchange structure on liquidity.

⁶ Indexed price is either the weighted average price of all trades in a day or the average of open, close, high and low prices.

⁷ For instance in April 2000, Yahoo's average quoted spread was 0.03% whereas average volatility was 0.39%.

The main results of this study are as follows. Institutional features of an exchange are the major determinants of liquidity in its listed stocks. After controlling for firm-specific (individual stocks volatility, trading turnover, average depth, size etc.), market-specific, and country-specific differences, this paper finds that the exchanges' institutional features have significant explanatory power in determining the differences in exchanges' liquidity performance. Both quoted and effective spreads as well as volatility are higher in dealer emphasis trading mechanisms (DLR) than in pure electronic-limit-order-book (LOB) or hybrid mechanisms (HYB)⁸. In developed markets, HYBs have marginally lower average spreads than LOBs but the difference is large in the emerging markets. Certain exchange-design features such as lower tick sizes, the presence of a designated market maker, a consolidated limit order book, automatic trade execution system, and order-flow centralization⁹ are associated with lower bid-ask spreads. The liquidity enhancing role of designated market makers is more pronounced in emerging markets than in developed markets. As the three measures of liquidity – spreads, volatility and trading turnover – are likely to affect each other, a simultaneous system-of-equations model is estimated using two-stage least squares (2SLS) methodology. Spreads are directly related to return volatility but inversely related to market size and turnover on a global basis. Consistent with previous studies, spreads and volatility are significantly higher in the emerging markets compared to those in developed markets. Moreover, trading intensity, defined as the ratio of annual trading volume to market capitalization, is higher on exchanges with automatic trade execution, and a consolidated limit order book. Finally, higher spreads, that widen the transaction cost band, seem to lower the incentive for trading.

⁸ On DLR markets dealers' quotes are primary source of liquidity, on LOB markets investors' limit orders are primary source of liquidity. On HYB markets these two sources of liquidity compete with each other and designated market makers have obligations to maintain orderly market and execute orders from their own account when necessary.

⁹ Here a market is defined as consolidated if all trading within a country in a given stock goes through a single system/single venue. When same securities are traded on scattered markets/multiple exchanges, we call the market fragmented.

Apart from their academic interest, these results carry policy implications for lawmakers who want to increase fairness and efficiency in securities markets. This study is also important for companies, investors and exchange managers. Better institutional design can improve liquidity and improved liquidity can potentially reduce the cost of equity for listed firms. By identifying better institutional features, investors can reduce transaction costs and improve the profitability of their investments. With better institutional features, stock exchanges can become more competitive and attract more investors for trading, and more firms for listing their stocks.¹⁰

The remainder of the paper is organized as follows. Section I links the institutional design of an exchange to its performance and develops the hypotheses to be tested. Data and summary statistics are presented in Section II. Classification of trading mechanisms of different exchanges are also described in this section. The empirical methodology and results are contained in Section III. Section IV discusses robustness issues, the practical utility of the results, and some limitations of this study. Section V concludes.

I. Hypotheses Development

The primary function of a stock exchange is to provide liquidity in listed securities. The effectiveness of a stock exchange in performing this function can be affected by many of its institutional characteristics, which we also refer to as exchange-design features. The features analyzed in this paper include exchanges' trading mechanisms, presence of designated market makers, tick sizes, existence of a consolidated limit order book, consolidated versus fragmented market, ex-ante transparency of order flow, automated trade execution, and exchange ownership.

¹⁰ Previous literature has documented various instances where trading turnover is found to be very sensitive to trading costs and market structure. See for example Pagano and Stiel (1996) who document that in 1989, French order handling rules made block trades unattractive and majority of block trades in French stocks were executed anonymously on the London SEAQ-International exchange.

The economic environment, shareholders rights, enforcement of insider trading rules, and individual stock characteristics are used as control variables.

The performance of exchanges is gauged by quoted spreads, effective spreads, realized spreads, volatility, and trading volumes. Percentage quoted spread on a stock is the difference between the lowest ask and the highest bid price, divided by bid-ask midpoint. This is representative of trading cost for the investors because the public is guaranteed to be able to trade at least small amounts at these prices without bearing negotiation costs. A further refinement to this concept is percentage effective spread, which is twice the absolute difference between actual transaction price and contemporaneous bid-ask mid-point divided by bid-ask midpoint. This measure is widely used because it accounts for price improvement/dis-improvement in actual trading. We also compute the realized spread, which is twice the signed difference between the closing transaction price and the midpoint of bid-ask quotes at the close of next session. This represents the order processing cost, inventory costs and rents of the suppliers of liquidity, net of adverse selection costs resulting from unfavorable price movements after the trade. The use of percentage spreads instead of absolute spreads for each of the three measures makes the comparison between exchanges more sensible because percentage spreads are free of numeraire.

The theoretical predictions on the relationships of the attributes of a stock exchange with its bid-ask spread performance are discussed below. All null hypotheses are set up in such a way that their rejection implies that the institutional feature under consideration significantly impacts performance measures.

A. Trading Mechanisms

Liquidity can be provided on a stock exchange by a variety of trading mechanisms such as a pure electronic limit order book (LOB), a dealer emphasis system (DLR), a hybrid mechanism (HYB) or a periodic call mechanism (CALL). In a LOB, incoming customer orders are matched

based on price and time priorities and all unmatched orders are accumulated in a consolidated order book for subsequent matching. In DLR, the brokers supply liquidity by quoting bid and ask prices at which customers can trade with them. However, customers' orders may not get the fullest exposure in DLR systems. In this study, we define HYB mechanisms as exchanges that use a combination of LOB and binding dealer quotes by designated market makers. Finally, the CALL mechanism, also called 'price fixing mechanism', accumulates orders over a period of time and then batch processes them at a single price that would maximize volume. The pure CALL markets, which account for trading in less than 5% of the world market capitalization, do not have any spreads and are therefore excluded from our analysis. Viswanathan and Wang (2002) predict that risk neutral investors prefer LOB markets but risk averse traders prefer DLR markets. Glosten (1994) theorizes that LOB provides maximum liquidity and therefore it is the optimal exchange-design. Parlour and Seppi (2001) postulate that HYB markets can compete and co-exist with LOB markets. We test whether the spreads are equal in the three categories against the alternative hypothesis that HYB has better liquidity as it uses two sources of liquidity.

H_{0} : Spreads are equal on quote-based dealer markets, limit-order books and hybrid mechanisms.

B. Presence of a Designated Market Maker

Some exchanges have designated market makers (MMs) who are obliged to supply bid and ask quotes and then act as counter parties for incoming orders by trading on their own account. The rationale behind having market makers is that they improve liquidity when the depth of the order book is not sufficient or lacks synchronization. However, Glosten (1994) predicts that LOB provides as much liquidity as possible and any additional competition (another exchange) is either unprofitable or redundant. An interpretation of this is that MMs do not add liquidity beyond that provided by a LOB system. He shows that no trader is worse off and many are strictly better off with an open LOB than with a monopolist specialist. Similarly, Black (1995) predicts that with automated

LOBs, dealers will lose money and therefore exchanges will have no market makers. Rock (1989) also suggests that market makers disrupt trading against the LOB and induce second order adverse selection. In a more recent paper, Stoll (1998) envisages that competition across markets reduces willingness of MMs to stabilize markets and electronic trading reduces the importance of MMs even in dealer markets. Thus, many theoretical models predict little role for a MM. On the other hand are models provided by Glosten (1989), Seppi (1997), and Parlour and Seppi (2001) who envisage that a hybrid specialist/limit order market provides better liquidity to small investors than the pure LOB. In these models, the specialist undercuts the public limit orders due to price and public priority, and thus lowers the transaction costs for small investors. Empirical evidence on the positive incremental value of specialist (designated market maker) is gathered in the context of options market by Mayhew (2002) and Anand and Weaver (2001). We test the hypothesis about contribution of market makers first for the whole sample and then separately for developed and emerging markets:

H1₀: Spreads are same on exchanges with market-maker and those without. As a result, spreads are also not different on hybrid markets and pure limit order books.

C. Tick Size

The theoretical models in Seppi (1997) and Parlour and Seppi (2001) show that tick sizes can significantly alter the dynamics of spreads faced by the investors in different institutional settings. Bacidore (1997) and Chakravarty, Harris and Wood (2001) show that bid-ask spreads reduce significantly as a result of reduction in tick size on Toronto and New York stock exchanges respectively. We test the null hypothesis that:

H2₀: Tick Size has no impact on spreads.

D. Consolidated versus Fragmented Markets

Biais (1993) envisions that the mean spreads are equal in both fragmented and consolidated markets but more volatile in the latter. Fragmentation potentially has two opposite effects. On the

one hand, it increases competition by increasing the number of dealers, which in turn reduces transaction costs. On the other hand, it splits the trading volume across trading venues and decreases price competition between orders thus decreasing liquidity (Madhavan 1995). Branch and Freed (1977) show that the first effect dominated whereas a NYSE (1973) study shows the opposite. Hamilton (1979) shows that the competitive effect exceeds the order-flow disintegration effect. In a related study, Arnold et al (1999) empirically show that mergers of US regional exchanges lowered bid-ask spreads. This can be interpreted to imply that too much fragmentation is bad. The issue is still unsettled in the literature. Here the null hypothesis is:

H3₀: Spreads are equal in both fragmented and consolidated markets.

E. Ex-Ante Transparency of Market Depth

Madhavan (1995) predicts that dealers in less transparent (opaque) markets will price more aggressively in early rounds to attract informed traders. The information learned can be used in later rounds to extract profits. In more transparent markets, dealers have no such incentive or opportunity. Pagano and Roell (1996) predict the opposite i.e. increases in both ex-ante and ex-post transparency lower spreads because it reduces the adverse selection problem for the dealers. Flood et.al (1999) and Bloomfield and O'Hara (1999) show in their experimental studies that an increase in transparency of quotes and trades widens spreads especially at the open or before news. However, in their studies, the differences in spreads disappear near the close of the trading round. Transparency can refer to many different information items and it can be ex-post or ex-ante. In this study an exchange is classified as transparent if the complete details of limit orders and quotes are displayed on the brokers' screens ex-ante.¹¹ Hence, the results have to be interpreted in this restricted sense.

H4₀: Closing spreads on transparent trading systems are equal to those on opaque systems.

F. Automatic Execution of Trades

¹¹ This essentially gives a snapshot of the demand and the supply schedules for the stock.

Domowitz and Stiel (1999) suggest that automation substantially reduces both the fixed and the variable costs of providing transaction services. There are tremendous savings in market development costs, distance costs, and order-processing costs when one compares automated systems with the non-automated ones. Coppejans and Domowitz (1997) show that the adverse selection components are also 17% higher on the CME floor, compared to the automated Globex trading system for stock index futures. Venkataraman's (2001) finding that spreads are wider on the automated Paris Bourse compared to floor based NYSE departs from the generally favorable view about automation. However, the paper acknowledges that it is very difficult to control for differences in insider trading laws, competition for order flow, and trading volume between United States and France. The two exchanges also differ in other aspects such as presence of a designated market maker on the NYSE and none on the Paris Bourse, tick sizes, speed of execution, and ownership structure. This study attempts to empirically examine the incremental impact of automation by controlling for other differences in a regression framework:

H5₀: Exchanges with automatic execution of trades experience the same spreads as those with trade-time re-negotiations/ broker intervention.

G. Ownership of the Exchange

Domowitz and Stiel (1999) discuss the implications of exchange governance mechanisms. Exchanges were traditionally organized as mutual associations co-owned and operated by member-firm brokers and dealers. Recently the trend is towards incorporation of exchanges. There are also a few exchanges that are owned by government entities. One implication of de-mutualization achieved via incorporation or government ownership is that the interests of the shareholders/regulators dominate broker-members' interests. As a result, policies that increase exchange's profits, competitiveness or parity are implemented even if it means a reduction in members' spreads and

profits. Incorporation also makes it easier to raise more finances and make continuous technological upgradations.

H6₀: Spreads on incorporated or government owned exchanges are equal to those on exchanges owned by a mutual association of exchange members.

H. Legal and Economic Environment: Shareholder Rights and Insider Trading Laws

The legal environment is another important determinant of the performance of a stock exchange. Regulation is a double-edged sword. On the one hand, it provides customer protection, financial system integrity and market price integrity. The literature supporting prohibition on insider trading argues that insider trading increases the degree of adverse selection. This forces liquidity provider to widen the bid-ask spreads. Bhattacharya and Spiegel (1991) prove that in the absence of laws against insider trading, spreads would widen to the extent that markets will break down. On the other hand, regulation can impose its own costs. Manne (1966) states that a ban on insider trading would reduce the efficiency of the markets and would impede an effective way to compensate managers. Empirical evidence in Bhattacharya and Daouk (2002) suggests that enforcement of insider trading laws significantly improves liquidity and reduces cost of capital.

H7₀: Better shareholder rights and enforcement of laws have no significant impact on spreads.

I. Trading Turnover, Volatility, and other Control Variables

In his presidential address, Stoll (2000) relates spreads to individual firms' trading characteristics in the following cross-sectional regression for US stocks listed on NYSE:

$$s = a_0 + a_1 \log V + a_2 \sigma^2 + a_3 \log MV + a_4 \log P + a_5 \log N + e \quad (1)$$

where s is the stock's proportional quoted spread defined as (ask price – bid price)/ P , V is daily dollar volume, σ^2 is the return variance, MV is the log of stock's market capitalization, P is the stock's closing price, N is the number of trades per day and e is the error term. The rationale for these variables is based primarily on order processing and inventory considerations. Increments in trading

volume, average size and number of trades, and firm size increase the probability of locating a counter-party, and thereby reduce inventory risk. The stock's return-variance measures the risk of adverse price change of a stock added to inventory. Stock's price-level controls for the effect of discreteness and is an additional proxy for risk because low price stocks tend to be riskier. Stoll (2000) finds that the empirical relationship in (1) is very strong and over 60% of cross sectional variance in spreads in NYSE stocks is explained by the independent variables (Adjusted $R^2 = 0.6688$). For the reasons discussed above, trading turnover, log of market capitalization, and volatility of returns are included in the regressions as explanatory variables. Data on number of trades, however, is not available. In order to control for differences in price levels of the stocks in the sample, we have the relative tick size (tick size expressed as a percentage of price level) as an independent variable in the regressions.

Note, however, that trading turnover and volatility are also dimensions of liquidity. Hence, we use these as additional performance measures in our paper. Both the variables, together with spreads, are used as endogenous variables in structural form equations following Domowitz et al (2002). Following Roll (1988), higher volatility is considered bad (a sign of illiquidity) and higher trading turnover is considered good (a proxy for deeper markets.)

Apart from these firm-specific control variables we also have several exchange-specific and country-specific control variables. The business environment in developed economies is also different from that in emerging economies. A dummy variable for developed countries is included in the regressions to account for these differences. Alternatively, in the robustness section, we also estimate a fixed effects model to control for any remaining country specific differences that are not captured by other variables. Further, performance of a stock exchange can improve or deteriorate with time. For that reason we include the age of a stock exchange as another control variable. Addition robustness checks are done with variables that measure differences in accounting

disclosure and reporting standards, ownership concentration of firms, and quoted depth across exchanges.

II. Data and the Details of Institutional Features

The hypotheses developed in the previous section address multiple issues in the institutional design of stock exchanges. Even though the different exchanges in the US itself provide some diversity in institutional features, the cross-sectional variation across the world markets appears much more promising to test these hypotheses in a unified framework. Accordingly, we have assembled the data on institutional features of 51 stock exchanges for which data on bid-ask spreads are available from the Bloomberg Financial Services archives and NYSE's Trades and Quotes (TAQ) database. These account for roughly 36% of the total number of exchanges in the world. In terms of market capitalization, however, the stock exchanges in the sample represent over 90% of the universe. The daily closing bid ask spreads are analyzed over the period from January 2000 to April 2000. Many market-microstructure studies use trade-by-trade data for one or two exchanges. Our study is much broader in its coverage of 51 exchanges but the data is restricted to daily closing bid price, ask price, and trade price for each stock. This information reasonably captures the trading costs faced by the investors as discussed in sub-section *E* where we compare daily and intra-day estimates.

A. Institutional details

The details about the trading mechanisms and organization of different exchanges are collected from various sources including a detailed survey of each exchange, information on home pages of exchanges on the World Wide Web. The website of International Federation of Stock Exchanges, www.fibv.com, as well as www.yahoo.com has internet links, phone numbers, fax numbers, and email addresses for all stock exchanges. In addition, directories, handbooks, and reports of capital market institutes like the International Financial Review's (1997-2003) handbook

of world stock & commodity exchanges, the Saloman Smith Barney's Guide to World Equity Markets, and Morgan Stanley Capital International's handbook of world equity and derivative exchanges are also used.¹² Bloomberg Financial Services also provides a brief profile of exchanges around the world. Wherever the required information was not available in any of these resources, it was obtained through direct correspondence with the exchange officials. The broad market data and classifications of markets into developed and emerging are obtained from Morgan Stanley Capital International.

B. Continuum of Institutional Design and Classification of Exchanges

The stock exchanges around the world display a continuum of possible institutional design features. For instance, the trading mechanism can be pure consolidated limit order book (LOB), hybrid system with an order book and designated market makers, or a dealer-quote driven system. Within the dealer-quote driven system some exchanges provide exposure to customer limit orders whereas others do not. Similarly ex-ante transparency can vary anywhere between completely opaque to completely transparent order books. In Appendix 1(a) to 1(f) we discuss the continuum of institutional design features, provide some examples of exchanges that have these features, and define the cut-off points for our discrete classification of exchanges on the following aspects:

- i. Designated Market Maker: The indicator variable 'market-maker' is set to one if the exchange employs one or more designated market makers (MMs) who are obliged to provide binding bid and ask quotes for some minimum quantity that they are ready to trade at all times.¹³ This variable is '0' if there are no MMs or only voluntary MMs. (Appendix 1.a.)
- ii. Trading mechanism: This can be Pure LOB if there is a consolidated limit order book within the exchange and no market makers. We define the trading mechanism to be hybrid, if in

¹² In about 74% of all data items on institutional design, the information was cross checked from two or more sources i.e. handbook of stock exchanges, websites, and direct emails from exchange officials.

addition to the consolidated order book, there are designated market makers for each stock. We classify the remaining exchanges as dealer-emphasis systems. Dealer emphasis systems are primarily dealer-quote driven, although they may have some provision for automated screen based trading. (See Appendix 1.b.) This classification is conservative in the sense that if we classify dealer system with screen based trading as limit order or hybrid systems, the gap in trading costs on dealer system and LOB/Hybrid system will widen even further.

- iii. Consolidated vs Fragmented Trading: Trading is said to be consolidated (fragment=0) if all domestic trades in any stock in the country are executed at a single venue or passes through a single execution system. On the other hand, if the same stock can be traded on multiple trading venues within the country, we classify the market as a ‘fragmented’ market (fragment=1.) Note that some stocks listed on domestically consolidated systems may be cross-listed on foreign exchanges and may have the status of domestically consolidated but globally fragmented. (Appendix 1.c.) However, as discussed in the robustness section of the paper, stocks that are cross-listed and other stocks on same exchange that are not cross-listed do not have statistically significant differences in spreads.
- iv. Transparency: If the details of the order flow like price schedules on the demand side as well as the supply side (bid depth for each price and ask depth for each price) are displayed on brokers’ screen then “ex-ante transparency” is set to 1. (Appendix 1.d.)
- v. Automatic execution: This is equal to one if trades are executed automatically based on price/time priorities or if the trades can be executed by hitting dealers’ quotes on the screen¹⁴. This indicator is 0 if some broker intervention is needed to complete the trade (Appendix 1.e.)

¹³ In practice market makers can avoid trading under very exceptional circumstances

¹⁴ For example the small orders execution system (SOES) on Nasdaq.

- vi. Ownership of Exchange: 'Owner'=1 if the stock exchange is owned and managed by a mutual cooperative of broker-members. If the exchange is incorporated or is an independent government organization then 'owner'=0.
- vii. Shareholders Rights and Insider Trading: La Porta et. al (1998) generate an index for shareholders' rights for different countries on a scale from 0 to 5 with zero representing least rights and 5 representing most rights. Bhattacharya and Daouk (2002) give information in their Table II about the date of the first enforcement of insider trading laws, which we use to determine whether or not the insider trading laws are enforced in a country. In our study we adapt La Porta's index by adding 1 to it if the insider trading laws are enforced in the country. Thus our proxy of shareholder rights has a scale of 0 to 6 with six being the best in terms of shareholders' rights.

C. Control Variables

We use Morgan Stanley Capital International's (MSCI) classification of markets as developed (=1) or emerging (=0). In the robustness section we include 51 dummy indicator variables for each exchange in the sample to estimate a fixed effects models. The remaining market-specific and firm-specific variables are continuous variables. The ages of all stock exchanges are computed from their establishment year to the year 2000. The total market capitalization (in billions of US dollars) for each exchange is the sum of market capitalization of all firms listed on that exchange. In regressions, the logged value of this market capitalization is used. The turnover figure presented is the ratio of annual value of total trading on the exchange to the total market capitalization of listed companies. We also gather the number of firms listed on each exchange in the year 1999. Three additional variables are used in robustness analysis. These are accounting standards, ownership concentration, and quoted depth. Accounting practices around the world are obtained from a survey conducted by Center for International Financial Analysis and Research Inc. The survey tells us how

strict are the disclosure and reporting requirements in 44 countries¹⁵. It gives details about reporting of consolidation of subsidiaries' accounts, hidden reserves, long term leases, deferred taxes, discretionary reserves, notes to accounts, major shareholders and changes in their holdings, shares held by directors/ employees and changes thereof, language of statements, big 6 or other auditors, and other balance sheet items. We construct an accounting standards index in the spirit of La Porta et al (1998) by adding one for each item disclosed by firms in a given country. The highest possible value of the index is 50. The actual values range from 12.5 for Greece to 39 for the US. Higher disclosure and frequent reporting reduce the information asymmetry for liquidity suppliers. Therefore, this variable is expected to have a negative coefficient in the regression with spreads as dependent variable.

Ownership structure of the firms is computed using the methodology followed by Pinkowitz, Stulz and Williamson (2003). Ownership concentration is defined as percentage of shares held closely by insiders. Insiders are considered to be officers, directors, and their immediate families, shares held in trusts, shares held by another corporation (except shares held in a fiduciary capacity by financial institutions), shares held by pension benefit plans, and shares held by individuals who hold 5% or more of the outstanding shares. Ownership concentration ranges from 8% for US firms to 78.1% for Czech Republic. Greater concentration of ownership reduces the float and trading turnover in the market. Therefore, this variable is expected to have a positive coefficient in the regression with spreads as dependent variable.

Data on quoted depth for the best bid and offer prices are not archived by Bloomberg Financial Services. Nevertheless, they provide current data on bid depth and ask depth for 33 exchanges. Therefore, for robustness checks we gather data on depths at the best bid and offer at the

¹⁵ Actual Survey had 44 countries. Remaining 11 countries were assigned average values for the region. For example all east European countries with missing data were assigned the average value of East European bloc.

close of each day in the week of December 17-21, 2001. Average depth for a stock is computed by adding the bid size and the ask size. Then we compute average depth for each exchange. The association between spreads and depths is important for this study. If exchanges with wider spreads are also able to offer bigger depths than then the spread measures in this study may not be sufficient to draw inferences about impact of exchange-design on liquidity. On the other hand if exchanges with wider spreads also have shallower depths than the inferences in this paper will be strengthened because both wider spreads and shallower depths imply lower liquidity.

D. Cross sectional variety in exchange-design

There is a large cross sectional variation between institutional design and set-up of exchanges around the world. More than 48,000 securities are listed on the 139 exchanges with an average of 345 companies per exchange. The aggregate market capitalization exceeds twenty five trillion dollars. The exchange with highest market capitalization is NYSE with more than \$8 trillion. The average annual trading turnover to market capitalization ratio is 0.749. Exchange age ranges from 1 year to 400 years; with oldest stock exchange being that in Germany.

The summary statistics for classification of stock exchanges based on their institutional features are given in Table I. In the sample of 51 exchanges, we have 20% dealer-emphasis markets, 51% pure limit order markets, and 29% hybrid exchanges. If we weigh the exchanges by their market capitalization, then dealer-emphasis system represents 23% market capitalization, pure limit order book account for 28% and hybrid markets 50%. In our sample, 63% of the exchanges operate in consolidated markets and the remaining in fragmented markets; 41% have full ex-ante transparency of order flow; 86% of exchanges have trading system with automatic execution of trades; Ownership of the exchange is in the hands of broker-members in 63% of cases; and 51% of the exchanges considered here operate in markets classified as developed markets by Morgan Stanley Capital International. The market-capitalization based classifications suggest that larger

exchanges are mutually owned, tend to have centralized and less than fully transparent, and enforce insider trading laws.

[Insert Table 1 here]

E. Performance Measures

Next, we compute the performance measures for 51 of these exchanges from bid, ask, and transaction prices at the close of each day. These data are collected from the Bloomberg Financial Services' archives and from NYSE's Trades and Quotes (TAQ) database. We identify the 25 stocks with highest market capitalization in December 1999 on each of the 51 exchanges using Datastream International. Data is not available for 56 stocks due to merger, acquisition, delisting or other similar reasons and we replace those with the stock having next highest market capitalization to get an initial sample of 1,275 firms. The selection is based on exchange of primary listing and therefore cross-listings through ADRs etc. do not get included. We do not apply any price filters that are commonly used in the US studies. Such filters will not have their desired impact in this study. For instance, if all stocks below US\$ 5 are excluded from the sample, then for many exchanges we will be left with no stocks to analyze. We still lose 4 firms from Estonia and 8 firms from Latvia as these countries only have a limited number of companies that are listed and for which quotes data at the close of the day is available in the sample period. This reduces the sample size to 1,263 firms. Since ADRs are not included, the sample has 1,263 unique firms with primary listing on the respective exchanges. Thus, there is no overlap due to cross listings. For these 1,263 stocks in the sample we gather closing quotes data from 1st January 2000 to 30th April 2000. These stocks represent on an average 75.80% of the total market capitalization of all stocks on each exchange.¹⁶

¹⁶ The mean, minimum, and maximum coverage are 75.80% for full sample, 22% for NYSE and 96.5% for Bermuda.

The five measures of performance used in the paper are quoted spreads, effective spreads, realized spreads, volatility of returns, and trading turnover. Table 2 gives the average relative tick sizes and the average of these performance measures across countries.

[Insert Table 2 here]

We use quote midpoints to avoid the bias in volatility computation due to bid-ask bounce. We also compute the variance of returns using last trade price. These numbers, not reported in the table, are very similar to the ones reported for volatility except for the high transaction cost countries. In particular, Estonia, Indonesia, Latvia and Ukraine have higher trading-price volatility than quote-midpoint volatility as the effect of bid-ask bounce is amplified.

In order to get rid of potential data errors we use the following filters to eliminate the day's observation if:

- i. Quoted spread is greater than 100% eg. bid is of \$ 10.45 ask is \$ 105.5 (in most likelihood 10.55 was incorrectly entered as 105.5)
- ii. Quoted spread is negative i.e. ask is less than bid (clear case of arbitrage).
- iii. Effective spread is more than 100% (this is same as (i) except that the error might be in transaction price.

These filters eliminate roughly 1.5% of all observations from the sample.

There is a lot of cross-sectional variation in the performance variables too. Table 2 shows NYSE has the lowest percentage quoted spreads (0.20%), lowest percentage effective spreads (0.10%) on the top 25 securities listed on the exchange. Ukraine has highest closing quoted (15.34%) and effective spreads (14.47%) followed by Bermuda.

There is an interesting pattern between quoted and effective spreads. Effective spreads are lower than or equal to the quoted spreads on 34 exchanges, which indicates that at least some trades are executed inside the quoted spread. However, on the remaining 17 exchanges, the effective

spreads are larger than the quoted spreads. This results when the quotes are only indicative but not binding and the prices are sensitive to actual trading. Usually, the quotes are binding only for small trade sizes. When trade size exceeds this minimum depth, the transaction price is likely to fall outside these bounds. The use of just closing (and not intra-day) bid prices, ask prices and transaction prices can potentially introduce measurement errors in effective spreads as there can be a lag between last trade and last quote revision. However, the use of highly active stocks from each exchange mitigates this problem to a large extent. For instance in the US, the quoted and effective spreads using closing data from TAQ are similar to those computed in previous studies that use intra-day data. Volatility of returns is lowest in Switzerland and highest in Ukraine for the top 25 stocks. Korea, NASDAQ, Taiwan and France have the highest trading intensity and Luxembourg has the lowest.

F. Comparison with other past empirical studies on spreads

The spreads for stock exchanges in developed markets are comparable to those in earlier studies like Perold and Sirri (1997) and Domowitz et al (2002).¹⁷ The correlation coefficients for quoted spreads, effective spreads, realized spreads, commissions, Roll's (1984) implied spreads, spreads documented in prior studies, and volatility are positive and remarkably high.¹⁸ They range from 40% to 97% and statistically significant at 1% level. Trading turnover is negatively correlated at approximately -40% with the spread measures. Thus the spreads in this study characterize the trading costs for investor in a realistic way.

The percentage spreads calculated in this paper for emerging markets are lower than those reported in the earlier studies, which might be a result of development of these markets.¹⁹

¹⁷ These studies report one-way costs which correspond to half of the 2-way round trip cost presented in this paper.

¹⁸ Roll's implied spreads are computed from the daily closing prices of the top 25 stocks and averages for each exchange.

¹⁹ Instead of computing cost for domestic investors, most of the earlier studies compute costs faced by a US institutional investor placing orders internationally. Such an investor might face higher market impact cost, greater intermediation costs etc. Moreover, the data used in this paper represent a more recent period when stock markets are developing at a fast pace in the emerging markets. This study also uses spreads on the 25 most active stocks in each market, whereas in

The quoted percentage spreads on top 25 stocks on these exchanges are also comparable with individual stock market studies. Stoll (2000) finds that the absolute quoted spreads on the largest market value decile's stocks are \$0.13 on NYSE and \$0.25 on NASDAQ. The numbers in this study are \$0.14 and \$0.34 respectively.²⁰ The 88 basis points spread for UK for the top 25 stocks compares to 104 basis point 'touch' computed in Hansch, Naik, and Viswanathan (1998). For the Paris Bourse, the spread is 0.396% for top 25 stocks close to 0.300% in Biais, Hillon, and Spatt (1995)²¹. Lehmann and Modest (1994) and Hamao, and Hasbrouck (1995) report average bid-ask spread of 0.817% and 0.83% respectively for the Tokyo Stock Exchange, both of which are comparable to 0.80% in this paper. Sandas (2001) presents the quoted spreads and average prices for the top 10 Swedish stocks trading on the Stockholm exchange. These convert to an average of 0.77% and the spreads in our study are 0.68%. Thus we can see that the closing bid-ask spreads in this study match with the current literature on international exchanges and thus these can be used as fairly reasonable estimates of trading costs that would be obtained from intra-day trading data.

III. Empirical Methodology and Tests

A. Descriptive Statistics

The sample period is divided into four monthly periods. Monthly values of average quoted spread, effective spread and volatility are created for each stock in the sample. These monthly values are then used as dependent variable. Monthly values are required to compute the volatility of stock returns because only one transaction price for each day is available. This procedure also reduces the measurement error due to random day-to-day fluctuation in spreads. Apart from other possible reasons such randomness is also induced by price discreteness resulting from tick size. Using daily

the previous studies, institutional investors might be investing in less active stocks. Finally, the implicit costs in previous studies are based on the difference between transaction price and a weighted average price. Here the computations are based on quoted bid-price, quoted ask price, and transaction price reported at the close of the market every day.

²⁰ Based on quoted spread of 0.20% and average price of \$69.79 on NYSE and 0.41% and \$82.83 on NASDAQ.

values can be problematic because if the model predicts spreads less than the minimum price variation (or any spread not falling on the specified price grid) on the exchange, then we may not observe such values. This averaging procedure has been widely used in the literature for example in Stoll (2000). Using this procedure, the initial sample of 89,460 daily observations results in a monthly sample size of 4,817 records (firm-months). This is 235 observations less than the 5,052 available firm months ($1,263 \text{ firms} * 4$) because of non-availability of data for some months and other data filtering previously discussed.

An unconditional comparison of average spreads across different market structures is presented in Table 3. Broadly speaking, the institutional structure of the market has a perceptible impact on the performance measures. Both spreads and volatility are highest in dealer-emphasis markets and lowest in hybrid markets. Spreads and volatility are higher on emerging markets compared to those on developing markets.

[Insert Table 3 here]

The differences in spreads described in panel A of Table 3 are statistically and economically significant²². In order to ensure that the differences are not being driven by concentration of one exchange type in developed or emerging markets, we conducted a two-way analysis by splitting the sample of exchanges between developed and emerging markets as shown in panel B and panel C respectively. The pure dealer markets in both developed and emerging economies have higher spreads and volatility compared to pure limit order books (LOB) or hybrid (HYB) markets. Next, we carry out a three-way analysis by splitting each of panels B and C into two sub-categories namely consolidated and fragmented locations of trading. The results of the three-way analysis are shown in Figure 1.

²¹ The price range of stocks is FF 100 to 500. We use the average of FF300 as denominator to calculate relative spreads.

²² The average spreads in table 3 are lower than the average in Table 2 because of Jensen's inequality. Table three is computed from monthly averages whereas table 2 is produced using direct averaging from daily data.

[Insert Figure 1 here]

The three market mechanisms are shown on z-axis, the different market classifications are on x-axis, and performance measures are on the y-axis. Again, in a variety of environments, hybrid markets (HYB) and pure limit order books (LOB) have lower spreads and volatility, and dealer-emphasis markets have the highest.

Finally, we need to account for the possibility that these differences may result from a combination of institutional features and individual stock trading characteristics. In order to simultaneously analyze the incremental impact of each institutional feature while controlling for firm specific characteristics, we need to perform a regression analysis. Furthermore, advanced econometric techniques also make it possible to model inter-dependencies between the performance measures.

Section I discussed a number of attributes of the stock exchanges that may in theory affect their performance in terms of spreads. In that section, endogeneity of spreads, volatility and turnover were also outlined. In order to account for the inter-dependencies in performance measures we need to use a simultaneous system-of-equations model. This application is developed in detail in the context of cross-country comparison of trading costs by Domowitz, Glen and Madhavan (2002). The endogenous variables are spreads, volatility and turnover. The exogenous or instrumental variables are the institutional variables and control variables. In addition to the theory discussed in Section I, we relied on Madhavan (1992), Madhavan (1995), Domowitz et al (2002), and Roll (1988) for the choice of explanatory variables for volatility and trading turnover.

The system of equations has three equations (equations 2,3, and 4) for the three endogenous variables i.e. spreads, volatility and trading turnover. The system is estimated three times using different measures of spreads i.e. quoted spread, effective spread and realized spreads (equations 2a, 2b, and 2c) is as follows:

$$pspread_{it} = \alpha + \beta_1 tick_{it} + \beta_2 mmkr_i + \beta_3 lob_i + \beta_4 frag_i + \beta_5 transp_i + \beta_6 auto_i + \beta_7 owner_i + \beta_8 mcap_{it} + \beta_9 develop_i + \beta_{10} right_i + \beta_{11} stdev_{it} + \beta_{12} trad_{it} + \varepsilon_{it} \quad (2a)$$

$$espread_{it} = \alpha + \beta_1 tick_{it} + \beta_2 mmkr_i + \beta_3 lob_i + \beta_4 frag_i + \beta_5 transp_i + \beta_6 auto_i + \beta_7 owner_i + \beta_8 mcap_{it} + \beta_9 develop_i + \beta_{10} right_i + \beta_{11} stdev_{it} + \beta_{12} trad_{it} + \varepsilon_{it} \quad (2b)$$

$$rspread_{it} = \alpha + \beta_1 tick_{it} + \beta_2 mmkr_i + \beta_3 lob_i + \beta_4 frag_i + \beta_5 transp_i + \beta_6 auto_i + \beta_7 owner_i + \beta_8 mcap_{it} + \beta_9 develop_i + \beta_{10} right_i + \beta_{11} stdev_{it} + \beta_{12} trad_{it} + \varepsilon_{it} \quad (2c)$$

$$stdev_{it} = \delta_0 + \delta_1 tick_{it} + \delta_2 mmkr_i + \delta_3 lob_i + \delta_4 frag_i + \delta_5 transp_i + \delta_6 auto_i + \delta_7 owner_i + \delta_8 develop_i + \delta_9 age_{it} + \eta_{it} \quad (3)$$

$$trad_{it} = \alpha_0 + \alpha_1 tick_{it} + \alpha_2 mmkr_i + \alpha_3 lob_i + \alpha_4 frag_i + \alpha_5 transp_i + \alpha_6 auto_i + \alpha_7 owner_i + \alpha_8 develop_i + \alpha_9 nfirm_i + \alpha_{10} pspread_{it} + \alpha_{11} stdev_{it} + v_{it} \quad (4)$$

where $pspread_{it}$ is the percentage quoted spread on security i in month t , $espread_{it}$ is the percentage effective spread on security i in month t , $rspread_{it}$ is the percentage realized spread on security i in month t , $stdev_{it}$ is the volatility of returns from security i in month t , and $trad_{it}$ is the trading turnover on the exchange i in month t . The independent variables are as follows. $Tick_{it}$ is the relative tick size expressed as a percentage of average price midpoint for each stock for each month, $mcap_{it}$ is the log of market capitalization (in millions of dollars) of the exchange in month t , $trad_{it}$ is the ratio of trading volume at an exchange to the market capitalization at that exchange, $nfirm_i$ is the number of firms listed on exchange i ; $mmkr_i$, lob_i , $frag_i$, $transp_i$, $auto_i$, $develop_i$, and $owner_i$ are indicator variables for presence of MM, limit order book, market fragmentation, transparency of order flow, automatic execution of trades, developed markets, and ownership of exchange by mutual cooperative of brokers. $Right_i$ is the index of shareholder protection laws and rights which we obtain by adding indices from La Porta et al (1998) and Bhattacharya and Daouk (2002). This ranges from 0 to 6 with 6 indicating highest shareholder rights. Age_{it} is the number of years since the establishment of the exchange. These variables are described in detail in Section II-B. ε_{it} , η_{it} , and v_{it} are the error terms.

The system of equations is estimated using the two stage least squares (2SLS) method. The number of exogenous variables excluded from each equation is at least as large as the number of

endogenous variables included in that equation. The model is also full rank. Thus, the rank and order conditions are met, implying that the system of equations is uniquely identified.

B. Parameter Estimates and Results of the Tests

The empirical relationship in equation (2a) for percentage quoted spreads gives an adjusted R-square of 31.91%. The adjusted R-square with percentage effective spread, as dependent variable is 30.77%. These regressions are based on a sample of 4,817 security months obtained from 89,460 daily observations. The regression coefficients are given in Table 4. This table reports the results for the system of equations using quoted spreads (equations 2a, 3 and 4). Partial results for the system using effective and realized spreads are also shown in the table (equations 2b and 2c). The corresponding numbers for volatility are identical as there is no simultaneity in that equation.

[Insert table 4 here]

After controlling for both individual stocks' trading characteristics like return variance, trading turnover and economy specific and firm specific characteristics like size, the institutional characteristics add significant explanatory power to explaining the differences in spreads across the exchanges. All the null hypotheses of equality are rejected at 1% level and, therefore, institutional design matters! If we rank the indicator variables for the institutional features by the magnitude of their coefficients, presence of consolidated limit order book, automatic execution of trades, and lower relative tick size have the maximum impact in reduction of spreads. Designated market makers lower trading costs whereas market fragmentation seems to be widening the spreads.

The coefficients on these variables are economically significant. For instance, presence of limit order book reduces the effective spreads by about 1.40%. This compares with the minimum percentage effective spread of 0.10% and average of 2.13% across all exchanges. Complete absence of limit order books in the world markets can cost the investors an extra \$210 billion on an annual

trading turnover of over \$15 trillion around the world. The impact on quoted spreads is even more dramatic.

The impact of lower relative tick size, fragmentation, transparency, and automatic execution respectively is also economically significant. Lower tick sizes result in lower spreads. Consolidated markets have lower spreads than fragmented ones after controlling for other differences in the regression. The effect of order-flow disintegration appears to be more harmful than the gains from increased broker competition in fragmented markets. Excessive ex-ante increase in transparency of details of order flow widens spreads consistent with Bloomfield and O'Hara (1999) and Ackert et al (2001). Automation is associated with reduced spreads. Results on ownership are mixed. Univariate statistics suggest that incorporated exchanges have better liquidity but regression results suggest otherwise. Trading intensity is higher on incorporated exchanges. The ownership structure may also matter in areas other than spreads like security innovation, and technology adoption. Better shareholder protections and enforcement of insider trading laws are associated with lower trading costs.

In line with extant literature, proportional spreads increase in return variance, and decrease in market capitalization and trading turnover. The 2SLS results also show that volatility of returns is lower in LOB systems. It increases with higher market fragmentation. Volatility is higher on newer exchanges and lower on older exchanges. Trading volume varies significantly with most of the market characteristics in the expected direction. Higher spreads widen the transaction cost band and lower the incentive for trading. Trading intensity is significantly higher in the emerging markets. Automation and aggregation of orders in a consolidated limit order book increases trading activity.

Several additional regressions were carried out by introducing interactive variables.²³ The results are qualitatively similar and some new insights are obtained. When we introduce interaction

²³ Full numerical results are not shown here for brevity but can be obtained by requesting the author.

between market maker dummy and development dummy, it is apparent that presence of market makers is more important for emerging markets (more negative coefficient) than it is for developed markets. This result has important implications for stock exchanges in emerging markets, which have abundantly adopted the pure limit order book model from the Paris Bourse, even though this system may not be optimal for less liquid securities. Assigning designated market makers to securities in addition to the electronic limit order book appears to offer definite gains in liquidity in emerging market stocks.

Thus we can conclude that institutional design and policies of stock exchanges potentially induce significant changes in trading costs. Investors appear to benefit by trading on exchanges with particular institutional features. In the next section we test the validity of this claim with further robustness checks.

IV. Robustness Checks, Limitations, and Practical Utility of the Results

Qualitatively similar results are obtained when different specifications and sub-samples are analyzed. These include OLS regressions for equations 2 to 4 with White's heteroskedasticity correction, regressions excluding the endogenous variables, sub-sample of only the top 10 stocks from each exchange, a fixed effects model with an intercept and 50 dummy variables to represent the 51 exchanges, separate month-wise regressions for the four sample months, separate region-wise regressions for developed and emerging markets, regression with 51 observations only by averaging across exchanges²⁴, sub-samples excluding the most liquid and the least liquid exchanges (possibly outliers), sub-samples excluding NYSE, sub-samples with marginal exchanges reclassified²⁵, and sub-sample of firms constituting the top 15% of market capitalization of each exchange. Although

²⁴ We average across exchanges to get rid of potential problems of correlated independent variables and autocorrelation. However, in this specification firm specific differences can't be controlled for and coefficient estimates are significant at 5% level instead of 0.01% level.

the coefficients vary slightly across the specifications discussed above, the R-square of regressions, and the direction and significance of the coefficients for the institutional variables are robust.

Cross-listing of stocks on international exchanges can affect the domestic market liquidity in a complex manner, balancing the costs of order flow migration against the benefits of increased inter-market competition. However, we find that the average spreads on each of the 51 exchanges for stocks that are listed only domestically and stocks that are cross listed on international exchange are not statistically different. We also test the impact of cross listing in the regression framework by including a indicator variable for global fragmentation which takes the value 1, if a stock is listed on more than one exchange anywhere in the world. The coefficient on this variable is statistically insignificant too.

The extensive use of dummy variables naturally gives rise to concerns about potential collinearity problems. We performed formal multicollinearity tests for the 5 regression equations. After scaling the matrix of explanatory variables ($\mathbf{X}'\mathbf{X}$) to correlation form, Belsley, Kuh, and Welsch (1980) construct the condition indices as the square roots of the ratio of the largest eigenvalue to each individual eigenvalue, $d_1 / d_j, j = 1, 2, \dots, p$. The *condition number* of the $\mathbf{X}$ matrix is defined as the largest condition index, d_1 / d_p . When this number is large, the data are said to be *ill conditioned*. A condition index ranging from 30 to 100 indicates moderate to strong collinearity. The raw conditional numbers for equations (2), (3), and (4) are 17.60, 13.80, and 16.42 and the intercept adjusted condition numbers are 2.95, 2.28, and 2.86 respectively which rules out severe collinearity.

Finally, we examine the association between quoted spreads and depths. Figure 2 plots the average quoted spreads and average depths for 33 exchanges for which current depth data was available from Bloomberg Financial services. This figure rules out a strong positive correlation

²⁵ For instance, some observers may argue that NYSE has fully transparent order-flow to the floor brokers. Therefore, we

between spreads and depths. Positive correlation between spreads and depth could have caused difficulties in interpreting the results because higher spreads indicate poor liquidity but higher depths indicate better liquidity. However, in reality exchanges with higher spreads seem to be having shallower depth as indicated by the downward sloping regression line. Thus both measures indicate a lower liquidity on such exchanges.

[Insert Figure 2 here]

The interpretation of results may also be subjected to endogeneity of the choice of exchange-design. For instance, if exchanges with lower liquidity endogenously choose dealer based mechanism than the analysis can be problematic. However, we argue below against this possibility. Emerging markets in general have higher spreads (lower liquidity) than developed markets. If exchanges with lower liquidity endogenously choose dealer based mechanism, then we should see a dominance of dealer based exchanges in emerging markets. However, the proportion of dealer-emphasis exchanges is roughly the same (20%) in both types of markets. Therefore, the current interpretation of results seems appropriate.

Some limitations of the study need to be borne in mind. Even though spreads and volatility are the most direct and relevant measure of trading costs for investors, there are other important criteria that investors might consider such as informativeness of prices. In addition, the results may be affected by the fact that spreads across exchanges are computed on different stocks that carry different levels of information and probably different inventory carrying costs. Moreover, the analysis applies to the stocks with very high market capitalization. These might well be the stocks with substantial investor interest. Nevertheless, the role of institutional features will be equally, if not more, important for smaller stocks. Broker commissions, inter dealer trading systems and policies for preferencing of trades may also differ across exchanges. To the extent possible, this

reclassify NYSE as a transparent exchange for robustness checks.

study tries to control for many of these differences indirectly by using control variables like market capitalization, economic development, shareholders rights, insider trading enforcement, and age of the exchange. We also look at accounting practices, ownership structure of firms, and quoted depth on the sample exchanges and the key results do not change significantly after allowing for these differences.

The practical utility of the results obtained in the previous section depends on several additional factors. When policy makers choose a particular aspect of institutional design, the performance measures discussed above may not be the only criteria. Clayton, Jorgensen, and Kavajecz (2000) find that a country's economic development, the degree of competition, extent of economic freedom, size of economy, availability of technology, and the legal system are important determinants of formation and structure (trading system) of international exchanges. However, in the last decade most exchanges have significantly altered their institutional features under competitive pressures. In the future also liquidity considerations are likely to emerge as the key drivers of institutional design of exchanges.

IV. Conclusions

This paper estimates the impact of institutional factors on the performance of 51 of the world's leading stock exchanges. To the best of our knowledge this is the first study documenting the actual average firm-level quoted and effective spreads from a large sample of stock exchanges. These estimates can serve as a good starting point for both practitioners and academicians concerned with transaction costs in cross-border investments. The paper also contributes to empirical literature on market microstructure by testing several theories that could not be tested using data from exchanges in a single country, where one may not observe the full spectrum of exchange designs.

After controlling for individual stocks' trading characteristics and several exchange-specific and country-specific differences, this study shows that the institutional differences help significantly

to explain liquidity differences across exchanges. In particular, dealer-emphasis systems have higher spreads and volatility compared to pure limit order systems (LOB) or hybrid systems (HYB).

Quoted, effective, and realized spreads are lower and trading volume is higher if an exchange has certain institutional features such a consolidated limit order book, designated market makers and provision of automatic trade execution. The role of designated market makers in reducing transaction costs is greater in the less liquid emerging markets compared to that in the more liquid developed markets. Large relative tick size and order-flow fragmentation within the country adversely affect trading costs. The institutional characteristics also affect volatility and trading turnover. Though these results are not conclusive, they serve to document several empirical relationships between the organization of stock exchanges and the bid-ask spreads on the exchange-listed securities.

This study can be extended in many ways. For example, the inclusion of smaller stocks in the sample can help analyze the impact of institutional design on liquidity of stocks with lower market capitalization. Likewise, this study focuses on how stock exchanges can vary their trading characteristics to become more competitive. It is assumed that lower spreads on listed securities will attract both more companies and more traders. However, future studies can examine additional factors influencing competitiveness such as market depth, price informativeness, broker commissions, transaction taxes, clearing fee, listing costs, and recurrent reporting costs and other dimensions of exchange-competition. Inter-dealer trading systems and order-flow preferencing policies also affect the exchanges' performance, and these too, can be analyzed.

Finally, even though this paper provides insights for government policy-makers, exchange owners and managers, brokers, companies, and investors, each of these players can view the implications of relationships studied here in a different way. Extensions to this study can focus exclusively on one such player or on interactions between them in dynamic settings.

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Appendix 1. Classification of the Markets around the world on the continuum of institutional features in Dec 1999

The continuum of possible features for six institutional aspects is given in six groups of rows. Examples of stock exchanges that have those features are given and the cut off points for discrete classification are marked with a vertical line within each group.

a.) Presence or absence of designated market maker

Classification	No Designated Market Maker =0		Designated Market maker =1
Continuum	Dealers can only act as agents for client orders. Don't trade as principal by law or custom.	Dealers can make market voluntarily but have no such obligation	Each stock is assigned to designated dealer(s) who is contractually obligated to continuously post binding bid-ask quotes within certain parameters
Examples	Latvia, Tokyo (Japan)	Australia, Denmark, Norway, Singapore Paris Bourse (France), South Africa New Zealand, Taiwan	Toronto (Canada), Italy, Ireland, AMEX, New York Stock Exch. (NYSE), Czech Netherlands, NASDAQ, EASDAQ, Poland

b.) Trading Mechanisms

Classification	Pure Limit Order Book	Hybrid Markets	Dealer Emphasis Market		
Continuum	Consolidated Limit Order book within the exchange and No market makers	Consolidated Limit Order Book within the exchange and Designated Market Makers	Dealer Quote Driven Markets/ Open-Outcry on Floor with some provision for electronic order book trading	Dealer Quote driven; Some exposure to customer limit orders	Pure Quote Driven Markets
Examples	Paris Bourse (France), Norway Australian, Denmark, Latvia Taiwan, South Africa, Singapore, New Zealand, Sweden, China German stocks in DAX index	New York Stock Exchange Toronto (Canada), Poland Ireland, Italy, Czech, German Stocks not in DAX	Brazil, London, Spain Indonesia	Nasdaq	Historical - NASDAQ Easdaq, Russian

c.) Centralized vs Fragmented Markets

Classification	Centralized =0	Fragmented =1
Continuum	Stocks trade on only 1 exchange/ system in the world	Stocks trade on more than 1 exchange/ system within the country
Examples	France, Australia, Greece, Canada	NYSE Stocks

Continued

Appendix 1continued

d.) Transparent vs Opaque Markets: Ex-ante Transparency of Order Book

Classification		Transparent =1			Opaque =0		
Continuum	Full order book is displayed to public	Details of best 5 bids and best 5 ask prices and depths are displayed to public at large	Full order book is displayed on brokers screen	Details of best 5 bids and best 5 ask prices and depths are displayed on brokers screen	Best Bid and Offer and corresponding depth only is displayed to public	Only best bid and offer displayed, No depths	Nothing is displayed
Examples	Island	Paris Bourse	Germany, Australia, Sweden, Singapore, Switzerland	Toronto (Canada), Denmark, Ireland	NYSE, Greece, Venezuela	Argentina, New Zealand	

e.) Automatic Execution vs Human Intervention

Classification		Automatic =1		Human Intervention =0	
Continuum	Orders are matched on price time priority automatically by a computer	Investors can hit dealers quotes for automatic execution		Orders are routed automatically, but human intervention takes place for execution	Completely manual matching
Examples	Paris Bourse, Singapore, Australia, Finland, Germany, HongKong, Ireland, Italy	NASDAQ-SOES, EASDAQ		NYSE Super-Dot, AMEX	Open-Outcry like Zimbabwe

f.) Ownership of the Exchange

Classification		Mutual Ownership =0		Mutual Ownership =1	
Continuum	Incorporated	A Government Department		Owned by Broker-members and Financial Institutions	Owned Solely by Broker-Members
Examples	Australia, Italy, Belgium, Bermuda, Denmark, Finland	Greece, Poland		Easdaq, Latvia	NYSE, Israel, Korea, New Zealand, S. Africa, Argentina, India, Malaysia, Mexico

Table 1. Classification of 51 exchanges by institutional characteristics

The first column describes the design feature, the second column counts the number of exchanges that have this feature, the third column divides this number by 51 to get the percentage, and the fourth column weights each exchange by its market capitalization to obtain the proportion of exchanges using that feature.

<u>Institutional Feature</u>	<u>Number</u>	<u>Percentage of 51 exchanges</u>	<u>Share of Market Capitalization</u>
Dealer Emphasis Systems	10	20%	23%
Pure Limit order Books	26	51%	28%
Hybrid (LOB+MM) Systems	15	29%	50%
Centralized Trading	32	63%	33%
Fragmented Trading	19	37%	67%
Full transparency of details of order flow	21	41%	27%
Opaqueness of details of order flow	30	59%	73%
Mutual ownership by brokers	32	63%	80%
Incorporated/ independent ownership	19	37%	20%
Developed Markets	26	51%	92%
Emerging Markets	25	49%	8%
Insider trading laws enforced	34	67%	96%
Insider laws not yet enforced	17	33%	4%
Total	51	100%	100%

Table 2. Average liquidity performance measures for 51 major exchanges.

We first calculate the liquidity measures for each stock for each day using the closing bid price, closing ask price and last trade price as follows:

(1) Relative Tick Size = Absolute tick size applicable to price range / Closing stock price

(2) Quoted percentage spread = (Ask Price – Bid Price) / Quote Midpoint.

(3) Effective percentage spread =

$(| \text{Transaction price} - \text{Quote midpoint} |) / \text{Quote midpoint} * 2.$

(4) Realized percentage spread = $\{ (\text{Transaction price} - \text{Quote midpoint}_{t+1}) / \text{Quote midpoint}_t * 2 \} * \text{inferred trade direction indicator}.$

where the subscript t indicates current trading day and t+1 indicates next trading day; inferred trade direction is assigned a +1 for buyer initiated trades and a –1 for seller-initiated.

(5) Volatility = Variance (Closing quote midpoint_t / Closing quote midpoint_{t-1})

(6) Trading Turnover = Annual Value of Total Trading in 1999/Total Market Capitalization in 1999

For each security the relevant liquidity measure is averaged across the sample period from 1st January 2000 to 30th April 2000. Finally, the averages over the 25 highest market capitalization stocks on each exchange are presented below.

	Relative Tick Size	Quoted spread	Effective Spread	Realized Spread	Volatility	Trading Turnover
Argentina	0.38%	1.74%	1.62%	1.71%	0.07%	0.236
Australia	0.16%	0.61%	0.56%	0.41%	0.08%	0.522
Austria	0.03%	0.85%	1.09%	1.00%	0.10%	0.377
Belgium	1.80%	1.00%	1.40%	1.20%	0.09%	1.446
Bermuda	1.07%	10.89%	7.93%	8.00%	0.08%	0.052
Brazil	1.16%	7.34%	5.34%	3.88%	0.38%	0.542
Canada	0.16%	0.54%	0.50%	0.69%	0.14%	0.571
China	0.09%	0.20%	0.22%	0.44%	0.18%	1.512
Colombia	0.15%	3.08%	2.73%	3.36%	0.13%	0.080
Czech	0.03%	3.66%	2.62%	2.96%	0.16%	0.347
Denmark	0.32%	1.50%	1.71%	2.00%	0.14%	0.579
EASDAQ	0.31%	6.12%	5.17%	4.55%	0.64%	0.078
Estonia	0.42%	4.53%	4.05%	5.10%	0.50%	0.140
Finland	0.12%	1.00%	0.95%	0.69%	0.19%	0.520
France	0.07%	0.40%	0.49%	0.58%	0.15%	2.555
Germany	0.02%	0.91%	0.77%	0.25%	0.11%	1.169
Greece	1.21%	0.74%	1.47%	1.63%	0.13%	1.216
Hong kong	0.41%	0.59%	0.56%	0.69%	0.15%	0.506
Hungary	0.20%	4.55%	2.48%	1.18%	0.11%	1.656
India	0.07%	0.81%	0.70%	1.24%	0.83%	0.657
Indonesia	1.70%	6.01%	6.51%	6.98%	0.56%	0.418
Ireland	0.32%	1.90%	2.39%	1.85%	0.13%	0.714
Israel	0.03%	0.95%	0.84%	0.55%	0.06%	0.393
Italy	0.15%	0.72%	0.92%	0.86%	0.14%	0.947

Continued

Table 2. Continued...

	Relative Tick Size	Quoted spread	Effective Spread	Realized Spread	Volatility	Trading Turnover
Japan	0.18%	0.80%	0.72%	0.71%	0.25%	0.494
Korea	0.18%	0.38%	0.37%	0.27%	0.30%	3.449
Latvia	3.98%	6.39%	5.48%	4.76%	0.30%	0.109
Luxembourg	0.31%	1.47%	1.68%	3.00%	0.25%	0.036
Malaysia	0.61%	0.96%	0.91%	0.77%	0.10%	0.343
Mexico	0.16%	1.76%	1.56%	1.50%	0.15%	0.292
Netherlands	0.05%	0.48%	0.55%	0.75%	0.15%	0.757
NewZealand	0.47%	1.30%	1.22%	1.20%	0.07%	0.456
Norway	0.42%	1.36%	1.29%	0.88%	0.17%	0.980
Peru	1.30%	4.35%	3.47%	3.46%	0.12%	0.227
Philippines	0.88%	1.97%	1.99%	2.30%	0.23%	0.445
Poland	0.31%	1.15%	1.00%	0.65%	0.18%	0.457
Portugal	0.13%	0.65%	0.69%	0.34%	0.10%	0.700
Russia	0.52%	5.38%	6.44%	10.19%	1.46%	0.074
Singapore	0.54%	0.85%	0.82%	1.26%	0.16%	0.751
South Africa	0.02%	0.77%	0.71%	0.63%	0.12%	0.343
Spain	0.06%	0.51%	0.50%	0.40%	0.08%	0.603
Sweden	0.31%	0.68%	0.68%	0.34%	0.10%	0.836
Switzerland	0.10%	0.43%	0.43%	0.46%	0.06%	0.820
Taiwan	0.44%	0.35%	0.36%	0.45%	0.09%	2.886
Thailand	0.88%	1.17%	1.18%	0.37%	0.18%	0.781
UK	0.03%	0.88%	0.83%	0.60%	0.14%	0.567
Ukraine	1.31%	15.34%	14.47%	14.60%	1.99%	0.084
US-Amex	1.17%	1.78%	1.30%	1.24%	0.21%	0.504
US-Nasdaq	0.13%	0.41%	0.51%	0.26%	0.40%	3.030
US-NYSE	0.12%	0.20%	0.10%	0.34%	0.12%	0.746
Venezuela	0.17%	5.87%	6.61%	6.34%	0.31%	0.197
Mean	0.49%	2.32%	2.13%	2.15%	0.26%	0.749
Std. Dev	0.68%	2.95%	2.63%	2.82%	0.34%	0.761
Minimum	0.02%	0.20%	0.10%	0.25%	0.06%	0.036
Maximum	3.98%	15.34%	14.47%	14.60%	1.99%	3.449

Table 3. Panel A. Comparison of spreads, tick-size and volatility across exchange types

This table gives an unconditional comparison of average daily closing spreads and volatility on exchanges with different mechanism and institutional features. Liquidity measures are averages for the top 25 market capitalization stocks on each exchange. Spreads that are statistically significantly different from the corresponding alternative design-feature at 5% level are shown in bold. Panels B and C show developed and emerging markets separately.

	Quoted spread	Effective Spread	Realized Spread	Relative tick size	Volatility
Average	2.41%	2.20%	1.92%	0.49%	0.24%
Dealer emphasis systems	4.62%	4.13%	3.79%	0.46%	0.50%
Pure limit order books	2.05%	1.97%	1.68%	0.60%	0.19%
Hybrid (LOB+MM) systems	1.64%	1.41%	1.17%	0.32%	0.16%
No designated market maker	2.48%	2.29%	1.95%	0.59%	0.22%
Designated market maker present	2.30%	2.06%	1.86%	0.33%	0.27%
Details of order flow opaque	2.84%	2.67%	2.40%	0.51%	0.31%
Details of order flow transparent	1.78%	1.54%	1.23%	0.46%	0.13%
Broker intervention required	5.11%	4.66%	4.82%	0.66%	0.61%
Provision for automatic trade execution	2.04%	1.87%	1.53%	0.47%	0.19%
Incorporated Exchanges	1.80%	1.68%	1.60%	0.52%	0.15%
Mutually Owned Exchanges	2.78%	2.53%	2.11%	0.47%	0.29%
Order flow Centralized to 1 exchange	2.03%	1.76%	1.45%	0.49%	0.15%
Order flow fragmented across exchanges	3.06%	2.98%	2.72%	0.49%	0.39%
Insider trading Laws not enforced yet	4.30%	4.05%	3.59%	0.61%	0.37%
Insider trading laws enforced	1.55%	1.37%	1.18%	0.44%	0.19%
Emerging Markets	3.81%	3.38%	2.93%	0.69%	0.32%
Developed Markets	1.16%	1.15%	1.02%	0.31%	0.17%

Panel B. Exchanges in Developed Markets

Dealer Emphasis Systems	2.16%	2.02%	1.82%	0.21%	0.32%
Pure Limit order Books	0.93%	0.94%	0.82%	0.41%	0.13%
Hybrid (LOB+MM) Systems	0.91%	0.95%	0.87%	0.25%	0.14%

Panel C. Exchanges in Emerging Markets

Dealer Emphasis Systems	7.45%	6.55%	6.08%	0.75%	0.73%
Pure Limit order Books	3.06%	2.89%	2.45%	0.78%	0.25%
Hybrid (LOB+MM) Systems	2.81%	2.15%	1.67%	0.43%	0.20%

Table 4. Two stage least squares (2SLS) regression of simultaneous system of equations model

This table shows the results of following regression of the triangular system of equations with liquidity measures as endogenous variables exchange-design features as instruments:

$$\begin{aligned} \text{spread}_{it} &= \alpha + \beta_1 \text{tick}_{it} + \beta_2 \text{mmkr}_i + \beta_3 \text{lob}_i + \beta_4 \text{frag}_i + \beta_5 \text{transp}_i + \beta_6 \text{auto}_i + \beta_7 \text{owner}_i + \\ &\quad \beta_8 \text{mcap}_{it} + \beta_9 \text{develop}_i + \beta_{10} \text{right}_i + \beta_{11} \text{stdev}_{it} + \beta_{12} \text{trad}_{it} + \varepsilon_{it} \\ \text{stdev}_{it} &= \delta_0 + \delta_1 \text{tick}_{it} + \delta_2 \text{mmkr}_i + \delta_3 \text{lob}_i + \delta_4 \text{frag}_i + \delta_5 \text{transp}_i + \delta_6 \text{auto}_i + \delta_7 \text{owner}_i + \\ &\quad \delta_8 \text{develop}_i + \delta_9 \text{age}_{it} + \eta_{it} \\ \text{trad}_{it} &= \alpha_0 + \alpha_1 \text{tick}_{it} + \alpha_2 \text{mmkr}_i + \alpha_3 \text{lob}_i + \alpha_4 \text{frag}_i + \alpha_5 \text{transp}_i + \alpha_6 \text{auto}_i + \alpha_7 \text{owner}_i + \\ &\quad \alpha_8 \text{develop}_i + \alpha_9 \text{nfirm}_i + \alpha_{10} \text{spread}_{it} + \alpha_{11} \text{stdev}_{it} + v_{it} \end{aligned}$$

where spread_{it} represents the percentage quoted, effective, or realized spread on security i in month t in 3 separate equations, stdev_{it} is the volatility of returns, and trad_{it} is the trading turnover on the exchange i in month t . The independent variables are as follows. Tick_{it} is the relative tick size expressed as a percentage of average price midpoint for each stock for each month, mcap_{it} is the log of market capitalization (in millions of dollars) of the exchange in month t , trad_{it} is the ratio of trading volume at an exchange to the market capitalization at that exchange, nfirm_i is the number of firms listed on exchange i ; mmkr_i , lob_i , frag_i , transp_i , auto_i , develop_i , and owner_i are indicator variables for presence of MM, limit order book, market fragmentation, transparency of order flow, automatic execution of trades, developed markets, and ownership of exchange by mutual cooperative of brokers. Right_i is the index of shareholder protection laws and rights. The analysis is based on 4,817 firm months obtained from 89,460 daily observations between January and April 2000 from 51 exchanges. Coefficients that are significant at 1% level are shown with **.

Dependent Variable	% Quoted spread		% Effective spread		% Realized spread		Volatility		Turnover	
Adjusted R-Square	0.3191		0.3078		0.2555		0.1095		0.2154	
Instruments										
Intercept	0.0656	**	0.0533	**	0.0444	**	4.7125	**	0.2887	**
Institutional Design Variables										
Relative Tick Size	0.0073	**	0.0057	**	0.0060	**	0.4043	**	-0.0029	
Market Maker	-0.0032	**	-0.0045	**	-0.0052	**	0.4174	**	-0.0514	
Limit order book	-0.0201	**	-0.0140	**	-0.0102	**	-0.9725	**	0.0852	**
Fragmentation	0.0045	**	0.0078	**	0.0059	**	0.8419	**	0.2008	**
Transparency	0.0068	**	0.0055	**	0.0031		-0.1189		0.2728	**
Automatic Execution	-0.0066	**	-0.0035		-0.0103	**	-0.4514	**	0.3644	**
Mutual Ownership	-0.0035	**	-0.0055	**	-0.0073	**	0.0336		-0.1435	**
Control Variables										
Market Capitalization	-0.0025	**	-0.0015		-0.0007					
Developed market	-0.0135	**	-0.0107	**	-0.0090	**	-0.2210		-0.1301	**
Number of firms listed									0.0002	**
Age of Exchange							-0.0027	**		
Shareholder rights	-0.0015	**	-0.0019	**	-0.0020	**			-0.0031	
Interrelated Liquidity measures										
Quoted Spread %									-0.0314	**
Volatility %	0.0029	**	0.0038	**	0.0050	**			0.0149	**
Trading Turnover	-0.0167	**	-0.0210	**	-0.0166	**				

Figure 1. Comparison of spreads and volatility with a three-way classification of 51 exchanges

These figures compare liquidity on dealer emphasis markets, pure limit order books, and hybrid systems in different environments. Exchanges are categorized as developed or emerging based on MSCI definitions. Then within each category, exchanges are further classified on the basis of fragmentation of order flow. This gives four categories of on the x-axis. For each category, the performance of the 3 trading mechanism is compared using four market quality measures, namely, quoted spreads, effective spreads, realized spreads, and volatility.

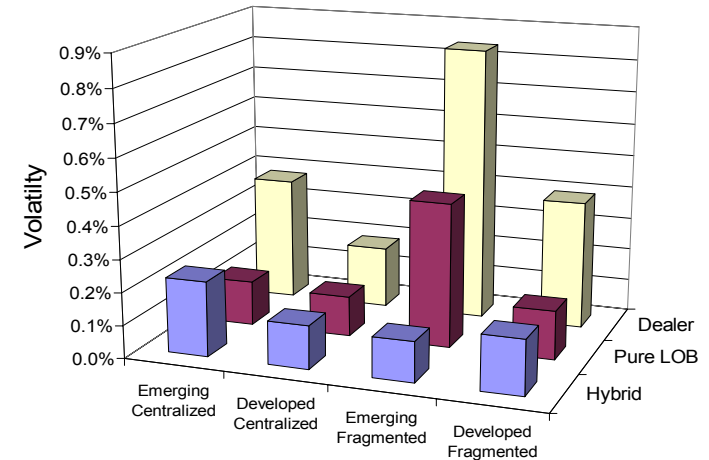
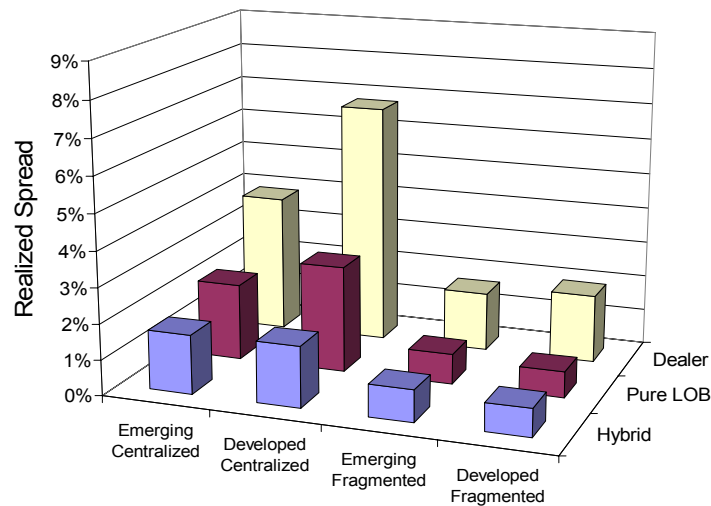
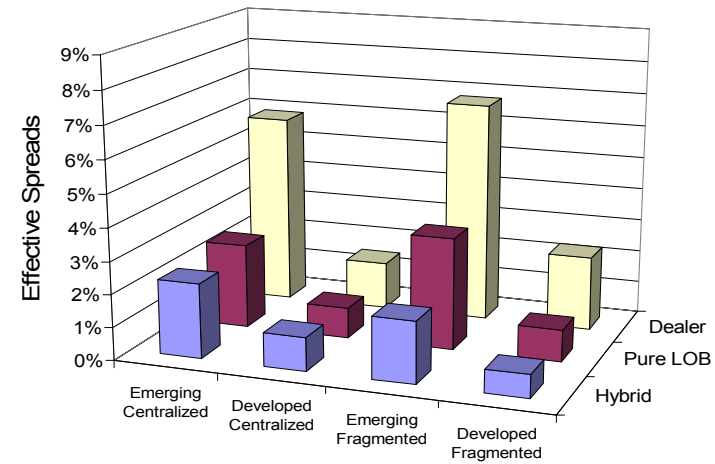
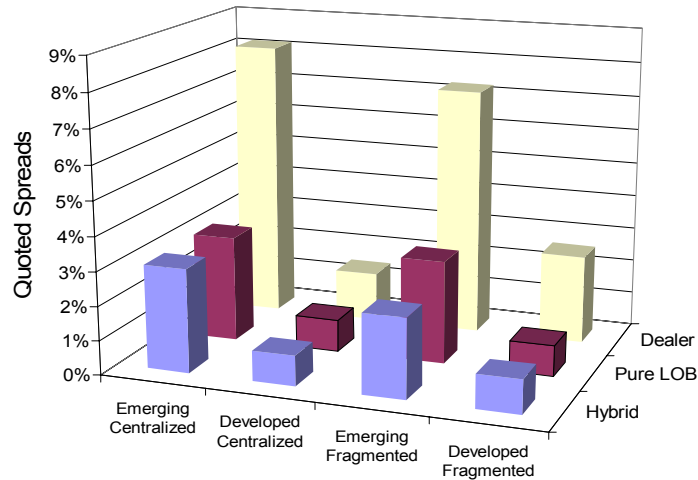


Figure 2. Quoted Spreads and Depths for 33 exchanges

This figure plots the average quoted spreads and average quoted depth for 33 countries for which the depth data is available at the close of each day from Bloomberg Financial Services. Depth in this chart is the log of sum of quoted depths at best bid and best offer. Scatter plot shows the actual depth-spread pairs representing averages across top 25 market capitalization stocks each in 33 countries. The thick dotted line in the middle represents the fitted regression with predicted values of depth plotted against the quoted spread values.

